Module: Computer and Programming Foundations

Phase: Foundations
Next module: Java Setup and First Programs

Module Outcome:

- Trace any app as **input** → **process** → **output** (roll number in, marks out)
- Tell hardware, OS, and a program apart — and why we write code at all
- Use bits, files, and a terminal well enough to survive JDK setup

What is a computer?

A **computer** is a machine that takes **input**, processes it using rules, and gives **output**. When you open a student portal on your phone, you type a roll number (input), the server checks the database (processing), and shows your marks (output).



Where used?
- College apps — student portal login, marks screens, lab machines
- **TechMart** / CampusCart — checkout flow (browser client → server → DB)
- **SecureBank** / UPI — ATM withdrawal; app request → bank server
- **AcmeHR** — employee tools that run on company laptops
- **MedCare** — appointment desks using folders, terminals, and web portals

Think of it like a kitchen:

```
[You give ingredients] → [Chef follows recipe] → [Dish served]
     INPUT                  PROCESSING              OUTPUT
```

What is an operating system (OS)?

The **operating system** is the manager of your laptop. Windows, macOS, and Linux are operating systems. The OS:

- Starts your laptop when you press the power button
- Shows the desktop and file explorer
- Runs programs (browser, VS Code, IntelliJ, `javac`)
- Manages files and folders on your hard disk

Without an OS, your laptop is just hardware with no way for you to interact easily.

What are files and folders?

A **folder** (also called a **directory**) is like a drawer in a cupboard. A **file** is a document inside that drawer.

Example structure for this course:

```
JAVA/
├── 01_Foundations/
```

```
├── 02_Core_Java/
└── projects/
    └── HelloJava.java
```

A **file path** is the full address of a file, for example:

- Windows: `C:\Users\Ravi\projects\HelloJava.java`
- macOS/Linux: `/Users/ravi/projects/HelloJava.java`

File extension (the part after the dot) tells the OS what kind of file it is:

Extension	Meaning
`.java`	Java source file
`.class`	Compiled Java bytecode (do not edit by hand)
`.txt`	Plain text
`.md`	Markdown notes
`.html`	Web page
`.json`	Data in JSON format

What is programming?

Programming is writing instructions in a language that a computer can follow. These instructions are saved in **source code** files.

An **algorithm** is a step-by-step plan to solve a problem — you can write it in plain English or on paper. **Code** is the algorithm written in a programming language like Java.

```
Algorithm (English):                Code (Java — coming in next module):
1. Ask user name                    // Scanner reads a line
2. Greet the user                   System.out.println("Hello, " + name);
```

You do not need to memorize code on day one. First understand the *problem* and the *steps*.

Client–server and the browser

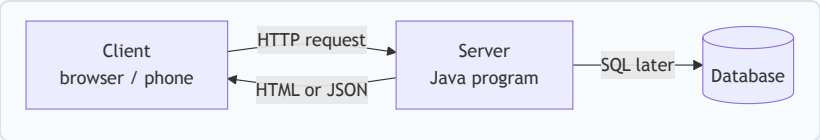
Most apps you use daily follow a **client–server** model:

```
[Client]  ---- HTTP Request ---->  [Server]
  (Browser / Mobile App)            (Java app on cloud)
[Client]  <--- HTTP Response ----  [Server]
              (HTML, JSON, data)
```

- **Client:** Your Chrome browser or phone app — it asks for data
- **Server:** A computer in a data centre (AWS, Azure, GCP) — it processes requests and sends replies
- **Request:** "Show me student ID 101's profile"
- **Response:** JSON or HTML page with the profile

When you open `https://example.com/login`, your browser (client) sends a request; the server checks credentials and sends back the dashboard page.

Visual — request and response:



Key takeaways:
- Client asks; server answers
- The browser does not store the company's database
- Same idea for Flipkart, UPI apps, and a college portal

Interview: "What happens when you type a URL?" → DNS finds the server IP → browser sends HTTP request → server returns HTML/JSON → browser renders.

Developer roles in India (entry-level view)

Role	What they build	Typical tools	Example in training
Frontend Developer	Screens users see and click	HTML, CSS, JavaScript, React	Student portal UI
Backend Developer	Logic, APIs, database	Java, MySQL	Employee login API
Full Stack Developer	Both UI and server	Java + React + MySQL	Expense tracker end-to-end
QA / Tester	Finds bugs before release	Test cases, Postman	Test login form
DevOps Engineer	Deploys and monitors apps	Git, Docker, AWS	Put app on server

In a 20-week full-stack program, you touch frontend, backend, database, and deployment — that is why the role is called **full stack**.

VS Code, IntelliJ, and terminal basics

VS Code and **IntelliJ IDEA Community** are free **IDEs** (Integrated Development Environment) — an editor with tools for coding, compiling, and running programs. Use **one** as your daily driver; both are fine for this course.

The **terminal** (also called command line or shell) is a text-based way to talk to your OS. Developers use it to compile Java, create folders, and use Git.

Important Concepts and Definitions

Term	Definition
Hardware	Physical parts: CPU, RAM, keyboard, screen
Software	Programs and apps that run on hardware
OS (Operating System)	System software that manages hardware and files
File	A named collection of data stored on disk
Directory / Folder	A container that holds files and other folders
Path	Location of a file or folder in the file system

Term	Definition
Algorithm	Ordered steps to solve a problem
Source code	Human-readable program text (`.java`)
Compiler	Translates source to bytecode (`javac`) — details next module
Client	Device or app that requests services
Server	Machine that provides services and data
HTTP	Protocol (rules) for web request/response
IDE	Editor with coding tools built in
Terminal / CLI	Command-line interface to the OS
Repository (repo)	Project folder tracked by Git (introduced later)

Step-by-Step Notes (With Examples)

Step 1: Know your machine

Check your laptop specs (Settings → About):

- **CPU:** Brain of the computer (Intel i5, Apple M1, etc.)
- **RAM:** Short-term memory while programs run (8 GB minimum recommended)
- **Storage:** Long-term space for files (SSD preferred)

> **Note:**
> - Wrong: Skipping OS / hardware basics and jumping straight to `javac`.
> - Prefer: One day on CPU, RAM, storage, and files — it saves weeks of confusion later.
>
> Write a short checklist note listing RAM, storage, and OS version you need for this course.

```
# My Laptop Specs — Priya (Hyderabad)

- OS: macOS / Windows / Linux
- CPU: _________________
- RAM: ____ GB (need ≥ 8 GB for IDE + browser)
- Storage free: ____ GB
- Ready for course? YES / NO
```

Create a markdown table classifying common laptop items as hardware or software.

```
| Item              | Hardware or Software? |
|-------------------|-----------------------|
| Keyboard          | Hardware              |
| Chrome browser    | Software              |
| SSD drive         | Hardware              |
| VS Code / IntelliJ| Software              |
| JDK 17            | Software              |
```

Step 2: Create your course folder structure

Open terminal and run (macOS/Linux):

```
# Create main project folder
mkdir -p ~/ksr_java_fullstack/projects
cd ~/ksr_java_fullstack
ls
```

Windows (Command Prompt or PowerShell):

```
mkdir C:\Users\YourName\ksr_java_fullstack\projects
cd C:\Users\YourName\ksr_java_fullstack
dir
```

Note:
- Wrong: Saving course files randomly on the Desktop; using spaces in folder names (`my project`).
- Prefer: One root folder (`ksr_java_fullstack`) and `snake_case` / `kebab-case` names (`my_project`).

Create the full course folder tree with notes, projects, and labs subfolders, then confirm your location with pwd and ls.

```
mkdir -p ~/ksr_java_fullstack/{notes,projects,labs}
cd ~/ksr_java_fullstack
pwd
ls
```

Expected Output: (paths vary by user)

```
/Users/you/ksr_java_fullstack
labs
notes
projects
```

Create a nested lab folder for day01_workspace and navigate into it.

```
cd ~/ksr_java_fullstack
mkdir -p labs/day01_workspace
cd labs/day01_workspace
pwd
ls
```

Step 3: Install and open the IDE

1. Download VS Code **or** IntelliJ IDEA Community
2. VS Code: install **Extension Pack for Java** (Microsoft). IntelliJ: bundled Java support.
3. **File** → **Open Folder** (VS Code) or **Open** (IntelliJ) → select `ksr_java_fullstack`
4. Create a file: `notes/day01_foundations.txt`

Note:
- Wrong: Opening a single file, then wondering why paths and terminals feel broken.
- Prefer: Always open the **project folder** as the root.

Write your Day 01 foundations notes covering what you learned about computers, the OS, and the terminal.

```
# Day 01 Foundations — Ravi

What I learned:
1. Computer = input → process → output
2. OS manages hardware and files
3. Terminal is how developers talk to the OS
```

Start a glossary with definitions for path, extension, and IDE.

```
# Glossary

- **Path:** full address of a file (e.g. `/Users/ravi/projects/HelloJava.java`)
- **Extension:** part after the dot (`.java`, `.md`, `.html`)
- **IDE:** editor with coding tools (VS Code or IntelliJ Community)
```

Step 4: Essential terminal commands

Command	macOS/Linux	Windows	Purpose
List files	`ls`	`dir`	See what is in current folder
Change directory	`cd folder_name`	`cd folder_name`	Move into a folder
Go up one level	`cd ..`	`cd ..`	Move to parent folder
Print working directory	`pwd`	`cd` (shows path)	Show current location
Create folder	`mkdir name`	`mkdir name`	New directory
Clear screen	`clear`	`cls`	Clean terminal view
Check Java (next module)	`java -version`	`java -version`	Verify JDK install

Annotated demo — navigate like a developer:

```
# You are here: /Users/priya
cd ksr_java_fullstack            # enter course folder
mkdir 01_Foundations             # create module folder
cd 01_Foundations                # go inside it
pwd                              # confirm path → .../01_Foundations
ls                               # list contents (may be empty)
```

Expected Output:

```
/Users/priya/ksr_java_fullstack/01_Foundations
```

(`ls` may print nothing if the folder is empty.)

Remember: `pwd` = “where am I?” — run it after every `cd` until paths feel natural.

Note:
- Wrong: Avoiding the terminal because it looks like “hacking” movies.
- Prefer: Run only the commands shown in class — `ls` / `cd` are safe navigation tools.

Practice navigating from home into ksr_java_fullstack/notes and confirm your path at each step.

```
cd ~
pwd
cd ksr_java_fullstack
mkdir -p notes
cd notes
pwd
ls
```

Drill moving up one directory level with cd .. and verify where you land.

```
cd ~/ksr_java_fullstack/notes
pwd
cd ..
pwd
ls
```

Common Error: Typing `cd ksr_java_fullstack` from the wrong folder → `No such file or directory`. Use `cd ~` first, or the full path `~/ksr_java_fullstack`.

Step 5: Algorithm vs code — student login example

Scenario: A college wants a simple login check.

Algorithm (pseudocode on paper):

```
START
  DISPLAY "Enter roll number"
  READ roll_number
  DISPLAY "Enter password"
  READ password
  IF roll_number is valid AND password matches
    DISPLAY "Welcome to student portal"
  ELSE
    DISPLAY "Invalid login"
  END IF
END
```

Flowchart (text form):

```
        [Start]
           |
    [Ask roll number]
           |
    [Ask password]
           |
    [Valid?]---No--->[Show error]--->[End]
       |
```

```
      Yes
       |
 [Show portal home]
       |
      [End]
```

You will write the Java version after Flow Control. Today, focus on *logic*, not syntax.

Note:
- Wrong: Mixing up *algorithm* (English steps) with *code* (Java syntax).
- Prefer: Write steps in plain English / pseudocode first; code comes later.

Write pseudocode for a SecureBank ATM withdraw flow that checks PIN and balance before dispensing cash.

```
START
  DISPLAY "Enter PIN"
  READ pin
  IF pin is correct
    DISPLAY "Enter amount (₹)"
    READ amount
    IF amount <= balance
      DISPLAY "Dispense cash"
      SET balance = balance - amount
    ELSE
      DISPLAY "Insufficient balance"
    END IF
  ELSE
    DISPLAY "Wrong PIN"
  END IF
END
```

Write pseudocode that reads marks and displays Pass or Fail based on a 35 cutoff.

```
START
  READ marks
  IF marks >= 35
    DISPLAY "Pass"
  ELSE
    DISPLAY "Fail"
  END IF
END
```

Step 6: Trace a real client–server flow

Example: Opening an **employee portal** in a browser.

1. Employee types URL → browser sends **GET request** to server
2. Server checks session / shows login page → **HTML or JSON response**
3. Employee submits form → **POST request** with username and password
4. Server validates against **MySQL database** → sends dashboard or error
5. Browser renders the page — employee sees payslip link

```
Browser                    Server (Java)                Database (MySQL)
   |                           |                            |
   |--- POST /login ---------->|                            |
   |    (emp_id, password)     |--- SELECT employee ------->|
```

```
    |                            |<-- row or empty -----------|
    |<-- 200 dashboard ----------|                            |
```

Note:
- Wrong: Treating client and server as the same kind of “computer” with no distinction.
- Prefer: Client = *your* browser; server = the company’s remote machine.

Label the client, request, server, database, and response for a campus canteen app.

```
# Campus canteen app (client–server)

- Client: Phone browser showing menu
- Request: GET /menu
- Server: Java program on cloud
- Database: MySQL table `menu_items`
- Response: JSON list of dosa, idli, chai with prices in ₹
```

Describe the expense tracker sync story showing why the server and database are the source of truth.

```
# Expense tracker sync

1. Ravi adds ₹120 chai on phone (client)
2. App sends POST /expenses JSON to server
3. Server INSERTs into MySQL
4. Friend Meera opens app → GET /expenses → sees Ravi's chai
Source of truth = server + database (not the phone alone)
```

Step 7: Write your career roadmap note

Create `notes/my_roadmap.md` in the IDE:

```
# My Java Full Stack Roadmap

- Goal role: Full Stack Developer (Java + REST + React)
- Timeline: 20 weeks
- Weeks 1–5: Core Java (types, flow, strings, arrays, methods)
- Weeks 6–9: OOP, collections, files, Git
- Weeks 10–12: MySQL + JDBC
- Weeks 12–16: Frontend + Spring Boot labs
- Weeks 17–20: React, testing, capstone, placement
- Portfolio target: CampusCart (college shop / booking)
```

Write a short note describing your target role, city preference, and first portfolio app.

```
# Role I want

Target: Full Stack Developer (Java)
City preference: Hyderabad / Bengaluru / remote
Why: Build both UI and APIs for campus/startup products
First portfolio app: Student marks CLI, then a small REST API later
```

Draft a weekly habit plan with study schedule and Naukri keywords to watch.

```
# Weekly habit (India IST)
```

```
- Mon–Fri: 1 live session + 45 min practice
- Sat: finish mini-project
- Sun: GitHub commit + update roadmap
- Naukri keywords to watch: Core Java, SQL, REST, Spring Boot, MySQL
```

Practice Exercises

Basic

1. Draw and label CPU, RAM, and storage on a blank laptop diagram.

Model answer: Sketch a laptop outline. Label **CPU** (processor — executes instructions), **RAM** (fast temporary working memory — apps currently open), and **storage** (SSD/HDD — permanent files, OS, projects). Mentally: CPU thinks, RAM holds today's desk papers, storage is the filing cabinet.

2. Create this folder tree using terminal only:

`ksr_java_fullstack/ ├── notes/ ├── projects/ └── labs/`

Model answer: From your Desktop (or training folder), run:

`bash mkdir -p ksr_java_fullstack/notes ksr_java_fullstack/projects ksr_java_fullstack/labs ls -R ksr_java_fullstack`

`mkdir -p` creates parent and child folders in one go. Confirm with `ls -R` that all three subfolders exist.

3. Write a 6-step algorithm for **withdrawing cash from an ATM** (no code).

Model answer:
1. Insert card / select account.
2. Enter PIN and authenticate.
3. Choose "Withdraw" and enter amount.
4. Check balance ≥ amount; if not, show insufficient funds and stop.
5. Dispense cash and update balance.
6. Print/show receipt and return card (or end session).

4. List five file extensions you expect to use in this course and what each is for.

Model answer:
- `.java` — Java source files
- `.class` — compiled bytecode (`javac` output)
- `.md` — Markdown notes / README
- `.html` / `.css` / `.js` — frontend pages (later)
- `.sql` — MySQL scripts (later)

Intermediate

1. On paper, draw client–server flow for an **expense tracker mobile app** syncing with a server.

Model answer: Phone app = **client**. Cloud/API + database = **server**. Flow: User adds expense → client sends HTTP request (JSON: amount, category, date) → server validates and stores in DB → server responds success/error → client updates local UI. Sync later: client pulls latest expenses from server on login/open.

2. Open the IDE, create `notes/glossary.md` with 10 terms from this module.

Model answer: Create `notes/glossary.md` with terms such as: CPU, RAM, storage, OS, algorithm, program, client, server, frontend, backend, terminal, IDE, full stack, HTTP, database. One-line plain-English definition each.

3. Compare two job posts on Naukri/LinkedIn for "Java Developer" (entry-level) — note required skills vs this course map.

Model answer: Typical entry posts ask: Core Java, OOP, SQL/MySQL, collections, Git, REST, Spring Boot, HTML/CSS/JS, sometimes React. Map to this course: Foundations → Core Java → OOP → Git → MySQL → JDBC → Frontend → Spring Boot → React → Testing/Placement. Soft skills (communication, problem-solving) appear often — practice explaining projects in plain English.

4. Write pseudocode for **task manager**: add task, list tasks, mark complete, exit.

Model answer:

```
tasks = empty list
LOOP forever
  show menu: 1 Add  2 List  3 Complete  4 Exit
  read choice
  IF choice == 1: read title; append to tasks as incomplete
  IF choice == 2: print each task with status
  IF choice == 3: read task number; mark that task complete
  IF choice == 4: stop
```

Interview-Oriented

1. Explain client–server model to a non-technical friend in 60 seconds.

Model answer: "Your phone is the **client** — it asks for data. The company's computer on the internet is the **server** — it stores and replies. Like ordering food: you (client) place the order; the kitchen (server) prepares and sends it back. WhatsApp chat, Flipkart pages, and banking apps all work this way."

2. What is the difference between an algorithm and a program? Give one example each.

Model answer: An **algorithm** is a human-readable sequence of steps to solve a problem (e.g. steps to calculate monthly salary after PF deduction). A **program** is that algorithm written in a language a computer can run (e.g. `Salary.java` that performs those steps).

3. Name three differences between frontend and backend with a real website example.

Model answer: Using Flipkart/Amazon-style shopping: (1) **Frontend** — product page layout, buttons, CSS you see in the browser; **backend** — stock check, price calculation, order save on the server. (2) Frontend runs mainly in the browser; backend runs on servers. (3) Frontend languages often HTML/CSS/JS/React; backend often Java + MySQL. Users see frontend; they don't see the database directly.

4. Why do developers use terminal when GUI (file explorer) exists?

Model answer: Many servers have no GUI. Git, `javac`, SSH, and deployment tools are CLI-first. Terminal commands are scriptable and repeatable. Interviewers expect comfort with `cd`, `ls` / `dir`, `javac`, `java`, and `git` — not only clicking folders.

Mini-Project (Assignment): Personal Developer Workspace Setup

Lab kit (build here): 15_Lab_Assets/01_terminal_workspace/

Goal: Leave the session with a repeatable workspace.

What to build

1. Create `ksr_java_fullstack` with subfolders: `notes`, `projects`, `labs`

2. Add `notes/day01_foundations.md` summarizing today's concepts in your own words
3. Add `notes/my_roadmap.md` with target role and 20-week milestones
4. Screenshot: IDE with folder tree visible + terminal showing `pwd` and `ls`
5. Submit screenshot + both markdown files (Google Drive / LMS)

Stretch

- Add `labs/day01_workspace/CHECKLIST.md` copied from the lab kit starter.
- Pin your course root folder in the IDE for one-click reopen.
- Write one Naukri/LinkedIn job note: 3 skills from this course that match an entry-level Java post.

Done when

- [ ] Folder tree `notes/`, `projects/`, `labs/` exists and opens in the IDE
- [ ] `notes/day01_foundations.md` and `notes/my_roadmap.md` are filled in your words
- [ ] Terminal shows correct `pwd` / `ls` for the course root
- [ ] Lab kit ticket steps completed

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. Explain what happens when you type a URL and press Enter.
A: Browser (client) sends HTTP request to server at that URL; server processes and returns HTML/resources; browser renders the page. DNS first resolves domain name to server IP.

Q2. What is the difference between hardware and software?
A: Hardware is physical (CPU, RAM); software is programs and data (OS, JDK, VS Code). Software runs on hardware.

Q3. What does a full stack developer do?
A: Builds both user-facing part (frontend) and server/database part (backend) of an application, and often deploys it — end-to-end product development.

Q4. Why learn terminal if we have an IDE GUI?
A: Servers often have no GUI; Git, `javac`, and deployment tools are CLI-first; terminal is faster for automation and industry standard.

Q5. Algorithm vs program — example?
A: Algorithm: steps to calculate employee monthly tax. Program: those steps written in Java and saved as `TaxCalculator.java` that the machine executes.